TOBE. ENERGY

COMPETITIVE LANDSCAPE REPORT

NON-CONVENTIONAL ELECTROLYSIS STARTUPS — BENCHMARKING ANALYSIS

— AUTHORIZED INVESTOR ACCESS ONLY

MARCH 2026

Sources: PitchBook company profiles, S&P Platts Hydrogen Price Index, company disclosures

SECTION 01

Executive Summary

The green hydrogen space has attracted billions in venture capital, but most funding has gone to companies iterating on conventional electrolysis architectures — PEM, alkaline, and SOEC. A smaller cohort of "non-conventional" startups claims breakthrough approaches.

This report benchmarks Tobe Energy against the three most comparable non-conventional electrolysis startups: **Advanced Ionics**, **Fourier**, and **HGen**.

> BOTTOM LINE
>
> Tobe Energy achieved a more advanced pilot system than all three comparables — with less capital, fewer employees, and full in-house engineering and manufacturing capability. While competitors are still scaling lab prototypes, Tobe is deploying commercially.

SECTION 02

Company Profiles

Advanced Ionics — Milwaukee, WI

Founded	2017 (8 years old)
Employees	26
Total Raised	$21.1M (Seed + $16.7M Series A, Nov 2024)
Pre-money Valuation	$42.3M (Series A)
Lead Investors	bp Ventures, Mitsubishi Heavy Industries
Technology	RSOEC-like vapor-phase electrolysis ("Symbion")
Operating Temp	~300°C
Membrane	None (vapor-phase eliminates traditional membrane)
Rare Earths	Nickel + stainless steel (no Pt/Ir)
Efficiency Claim	35 kWh/kg target (stack only, not demonstrated); ~75% HHV est.
Pilot Scale	800 cm^2 active area
Pilot Cost	~$4M
Manufacturing	Partnered scale-up (not in-house)
Status	Pre-revenue, R&D stage

> Founded in 2017 with $21M raised and 26 employees over 8 years. Still at 800 cm^2 lab scale. The vapor-phase approach has theoretical thermodynamic advantages but requires 300°C operating temperature, adding thermal management complexity that offsets efficiency gains at the system level.

Fourier — San Francisco, CA

Founded	2021
Employees	5
Total Raised	$23.6M (Seed + $18.5M Series A-1/A-2 + debt, Dec 2024)
Pre-money Valuation	$62M (Series A)
Lead Investors	Airbus Ventures, General Catalyst, Paramark Ventures
Technology	PEM stack (conventional architecture, "modular blade" form factor)
Membrane	Yes — standard PEM membrane (degrades)
Rare Earths	Yes — platinum, IrO_2
Efficiency	50–55% HHV (standard PEM range)
Pilot Cost	~$5M
Manufacturing	Assembly only (components sourced externally)
Status	"Generating Revenue" (PitchBook) — likely small pilot sales/grants

> Despite the "non-conventional" positioning, Fourier is essentially a conventional PEM electrolyzer repackaged in a modular form factor. Still requires platinum, iridium, and membranes. $62M pre-money valuation for 5 employees and a small modular PEM system.

HGen — Hawthorne, CA

Founded	2021
Founders	Molly Yang (CEO), Colin Ho (CTO)
Employees	6
Total Raised	$7M ($2M Seed Jul 2022 from Founders Fund + $5M Seed-1 Sep 2024 from Seven Seven Six)
Post-money Valuation	$25M
Lead Investors	Founders Fund (Seed), Seven Seven Six (Seed-1)
Other Investors	Breakthrough Energy, Climate Capital, Fontinalis Partners, Yishan Wong (fmr Reddit CEO)
Technology	Modular electrolyzers; SpaceX/Tesla-inspired manufacturing optimization; focuses on optimizing "the entire widget," not just the stack
Target Markets	Steel, ammonia, decarbonization
Cap Table	Seed-2 stock at $1.03/share (16.67% acquired); Seed-1 at $1.50/share (20% acquired); non-participating preferred, weighted avg anti-dilution
Status	"Generating Revenue" (PitchBook) — no public revenue figures

> SpaceX/Tesla manufacturing DNA with high-profile investors (Founders Fund, Breakthrough Energy, Ohanian). $25M post-money valuation on $7M total raised with 6 employees. Listed as generating revenue on PitchBook but no public figures. Focuses on full-system optimization rather than novel cell chemistry.

SECTION 03

Head-to-Head Comparison

METRIC	TOBE ENERGY	ADVANCED IONICS	FOURIER	HGEN
Founded	**2024**	2017	2021	2021
Age at comparable stage	**<1 year**	7 years	3 years	3 years
Employees	**9**	26	5	6
Total raised	**$1.8M**	$21.1M	$23.6M	$7M
Pre-money valuation	—	$42.3M	$62M	$25M (post)
Lead investors	Cortado Ventures	bp, Mitsubishi	Airbus, General Catalyst	776, Founders Fund
Electrolysis type	**Isothermal, low-temp, membrane-free**	Vapor-phase (RSOEC-like)	PEM (conventional)	Undisclosed
Operating temp	**<30°C**	~300°C	Unknown (high)	Unknown
Membrane	None	None (vapor)	PEM (degrades)	Unknown
Rare earths	None (304 SS)	Nickel + SS	Pt, IrO_2	Unknown
System efficiency (HHV)	**>92% (42 kWh/kg)**	~75% est.	50–55%	Unknown
Pilot active area	**6,400 cm²**	800 cm²	Small blade	Unknown
Pilot cost	**~$300K**	~$4M	~$5M	Unknown
Engineering scope	**Full-stack (system + cell + circuit + controls)**	Stack only	Off-the-shelf PEM	Unknown
Manufacturing	**Full in-house (CNC, PCB, transformer)**	Partnered	Assembly only	Unknown
Commercial deployment	**Q4 2026 — 600kW at Zeeco ARC**	Not yet	Not yet	Not yet
Published pricing (Platts)	Yes — $7.05/kg delivered	No	No	No

SECTION 04

What This Means

1. Capital Efficiency

Tobe built a 6,400 cm² pilot system for $300K. Advanced Ionics spent $4M for an 800 cm² system. Fourier spent $5M for a small modular blade.

COMPANY	PILOT COST	ACTIVE AREA	$/CM²
Tobe	**$300K**	**6,400 cm²**	**$47/cm²**
Advanced Ionics	$4M	800 cm²	$5,000/cm²
Fourier	$5M	~500 cm² (est.)	~$10,000/cm²

100x

MORE CAPITAL EFFICIENT

vs. Advanced Ionics on $/cm² basis

8x

LARGER PILOT AREA

6,400 cm² vs. 800 cm²

$300K

TOTAL PILOT COST

vs. $4M–5M for competitors

STRUCTURAL ADVANTAGE

Tobe is **100x more capital-efficient** at the pilot stage. This isn't an incremental advantage — it's a structural one that comes from vertical integration and in-house engineering.

2. Speed of Execution

- Advanced Ionics was founded in **2017** and is still at lab scale after 8 years and $21M.
- Fourier was founded in **2021** and has a small modular PEM unit after 4 years and $24M.
- Tobe was founded in **2024** and is deploying a **600kW commercial system** within 2 years of founding on $1.8M.

3. Technology Differentiation

Advanced Ionics' vapor-phase approach eliminates the membrane but introduces a 300°C operating temperature — requiring insulation, thermal management, and high-temperature materials that add cost and complexity. Fourier isn't really non-conventional at all — it's a PEM electrolyzer in a modular package, still dependent on platinum, iridium, and membranes.

Tobe operates at **<30°C** with no membrane, no rare earths, and no cooling system. The simplicity of the architecture is the competitive advantage.

4. Pricing Transparency

Tobe is the only company in this cohort to publish its hydrogen pricing methodology through an independent third party (S&P Platts). At $7.05/kg delivered with 30% margin and no subsidies, Tobe's pricing is verifiable and competitive with conventional hydrogen delivery.

SECTION 05

Timeline

2017
Advanced Ionics founded in Milwaukee, WI

2021
Fourier founded in San Francisco, CA; HGen founded in Hawthorne, CA

2024
Tobe Energy founded in Oklahoma City, OK

2024 (NOV)
Advanced Ionics raises $16.7M Series A — still at 800 cm² lab scale

2024 (DEC)
Fourier raises $18.5M Series A — small modular PEM blade

2025
Tobe raises $1.8M pre-seed, builds 6,400 cm² pilot, publishes Platts pricing

2026
Tobe deploys 600kW commercial system at Zeeco ARC — while all three competitors remain in lab/early R&D

SECTION 06

Key Takeaways

$1.8M	<2 yrs	>92%
TOTAL RAISED	FOUNDED TO COMMERCIAL	SYSTEM EFFICIENCY (HHV)
vs. $7–24M for comparables	vs. 4–8+ years for comparables	vs. 50–75% for comparables

THE CAPITAL EFFICIENCY STORY

"$300K pilot with 8x the active area of a $4M system. That's not luck — that's what happens when you build everything in-house."

THE DEPLOYMENT STORY

Commercial production is the only metric that ultimately matters. While Advanced Ionics iterates on lab prototypes after 8 years, Fourier repackages conventional PEM technology, and HGen remains in stealth — **Tobe is shipping a 600kW system to a paying customer.**

Sources: PitchBook company profiles (via Stanford GSB access), S&P Platts Hydrogen Price Index, company disclosures. March 2026.